Native Binarization: Training 1-Bit-Weight Diffusion Models from Scratch on MNIST

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***Abstract*—The memory footprint and arithmetic cost of floating-point weights are a central obstacle to running generative diffusion models on edge hardware. Binarizing weights to a single bit would help on both fronts, but a widely held view—articulated most recently by the Binarized Diffusion Model (BiDM) [1]—is that fully binarizing a diffusion network is *catastrophic*, and that recovery requires distilling from a full-precision teacher. In this paper we revisit that claim in a deliberately small and controlled setting: unconditional generation of 28×28 MNIST digits. Rather than treating binarization as a post-hoc compression step, we train a residual U-Net with 1-bit weights *from random initialization*—an approach we call Native Binarization—using mean-centered weight scaling and pre-activation residual blocks. In a controlled W1A16 comparison (binary weights with full-precision activations, so that only the binarization *methodology* varies), Post-Training Quantization (PTQ) does collapse on this benchmark (FID 191.94, legibility 64.92%), consistent with the catastrophic-failure narrative, whereas the natively trained model reaches FID 24.24 with 89.45% legibility—comparable to an FP16 baseline of identical architecture (FID 29.60, 90.79%). A fully binary W1A1 variant degrades but does not collapse (FID 83.98). We emphasize that these findings are confined to MNIST, whose data manifold is far simpler than that of natural images; we therefore present them not as a general solution to binary diffusion but as preliminary evidence that, at least on this benchmark, the collapse commonly attributed to 1-bit precision is better explained by the train-then-quantize pipeline than by binarization itself. All code and checkpoints are released to support replication.**

Keywords: 1-Bit Quantization, Diffusion Models, Model Compression, Binary Neural Networks, Post-Training Quantization, ResUNet, Edge AI, MNIST

I. Introduction

Denoising diffusion models have become one of the most capable families of generative models in machine learning. Denoising Diffusion Probabilistic Models (DDPMs) [2] produce high-quality images across many domains, but their reliance on floating-point matrix arithmetic ties them to power-hungry GPUs and server-grade accelerators. With growing interest in on-device intelligence—in mobile phones, wearables, and IoT sensors—this dependence is a practical concern: executing the hundreds or thousands of sequential denoising steps a typical DDPM requires is difficult on an edge processor today.

One appealing direction is to push weight precision toward its minimum: a single bit. Compressing 16- or 32-bit parameters into $\{-1,+1\}$ values would replace costly multiply–accumulate (MAC) instructions with XNOR-popcount logic, reducing both memory and energy substantially. A recurring conclusion in the literature, however, is that such extreme compression breaks diffusion models. The most recent study of this question—the Binarized Diffusion Model (BiDM) [1], presented at NeurIPS 2024—states plainly that "the impact of fully binarizing weights and activations is catastrophic," and builds an elaborate distillation pipeline from a full-precision teacher to recover usable quality.

This paper is a small, focused attempt to look more closely at *where* that catastrophe comes from. Our starting observation is that binarization has, in practice, almost always been applied as **Post-Training Quantization (PTQ)**: a carefully trained floating-point model has its weights snapped to binary values after the fact. This "train first, binarize later" pipeline discards the finely tuned magnitude relationships that encode the learned score field. We ask a simple question: if the weights are constrained to be binary *from the first gradient step*, rather than compressed after training, does the collapse still occur?

We study this question in a deliberately narrow setting—unconditional MNIST generation at 28×28 resolution—and refer to the from-scratch approach as **Native Binarization**. It treats binarization as a problem of *representation* rather than compression: starting from random initialization, gradient descent (via the straight-through estimator) searches for a configuration of $\{-1,+1\}$ weights that directly minimizes the DDPM loss. Two design choices support this: (1) **mean-centered weight scaling**, which preserves what we informally call *structural dominance*—the dominant directional structure of each filter is retained after binarization—and (2) **pre-activation residual ordering** (BN $\rightarrow$ Act $\rightarrow$ Conv), which we use to keep gradient flow through binary layers well-conditioned during training (*topological stability*). We frame both as working hypotheses that are consistent with our observations, not as proven laws.

Our central comparison is a controlled **W1A16** ablation: activations are held at full precision while only the weight-binarization method varies (PTQ versus native). On MNIST, PTQ degrades sharply (FID 191.94, legibility 64.92%)—the kind of collapse the catastrophic-failure view predicts—whereas the natively trained binary-weight model reaches FID 24.24 with 89.45% legibility, comparable to an FP16 baseline of the same architecture (FID 29.60, 90.79%). We are cautious about the direction of this small gap and do not claim the binary model is genuinely *better* than full precision; we read it as "no meaningful loss on this benchmark." A fully binary W1A1 variant (binary weights and activations) degrades to FID 83.98 but remains far from collapse.

Scope and contributions. We want to be explicit that this is a small-scale study on a single, simple dataset. MNIST digits occupy a far lower-dimensional manifold than natural images, and results here need not transfer to CIFAR-10, CelebA, or LSUN. Within that scope, our contributions are:

- We provide preliminary evidence, on MNIST, that a 1-bit-weight diffusion model trained *from scratch*—with no full-precision teacher, no distillation, and no pre-trained starting point—can match an FP16 baseline of identical architecture. This is a limited but concrete counterpoint to the view that distillation from a teacher is indispensable for binary diffusion.
- We articulate two informal design principles—**structural dominance** (centered binarization preserves directional weight geometry) and **topological stability** (pre-activation ordering keeps binary training well-conditioned)—that are consistent with why native training behaves differently from PTQ in our experiments.
- We construct a controlled **W1A16 ablation** that isolates the weight-binarization methodology from activation-quantization noise, enabling a direct PTQ-versus-native comparison at the 1-bit weight boundary on MNIST.
- We report a simple **Legibility Score**, a per-sample class-confidence metric that complements FID by measuring whether each generated digit commits to a recognizable class, and use it alongside FID across the FP16, W1A16, and W1A1 regimes.

II. Related Work

A. Diffusion Models and Edge Deployment

A DDPM [2] corrupts training images with Gaussian noise over T forward steps and trains a network f_θ to reverse this corruption, with loss

$$\mathcal{L} = \mathbb{E}_{x_0,\varepsilon,t}\Big[\big\|\varepsilon - f_\theta\big(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\ t\big)\big\|^2\Big] \qquad (1)$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$. Follow-up work—including DDIM [3], Latent Diffusion [4], and SDXL [5]—has improved image quality substantially, yet every variant inherits the same inference bottleneck: many sequential forward passes through the denoising network. Efforts to bring diffusion models to edge devices have largely targeted sampling speed (e.g. progressive distillation [6]) or network pruning; neither addresses the underlying cost of floating-point arithmetic itself.

B. Post-Training Quantization and Its Limits

PTQ takes an already-trained model and re-maps its floating-point parameters to a lower bit-width without retraining. Q-Diffusion [7] and PTQD [8] showed that diffusion models can be quantized to 4 or 8 bits with tolerable quality loss, and later methods added timestep-aware calibration to account for the fact that activation statistics shift across denoising steps. PTQ quality nonetheless degrades quickly as bit-width shrinks; at the 1-bit extreme, quantization error dwarfs the weight magnitudes. Our own MNIST experiments are consistent with this: a W1A16 model produced by PTQ records FID 191.94 and 64.92% legibility, whereas the same architecture trained natively under binary constraints reaches FID 24.24 and 89.45%.

C. The 1-Bit Binarization Bottleneck

In a Binary Neural Network (BNN) [9], [10], both weights and activations are restricted to $\{-1, +1\}$, turning multiply–accumulate arithmetic into XNOR-popcount logic. XNOR-Net popularized per-channel scale factors ($\alpha = \text{mean}|W|$) to narrow the quantization gap, and the Straight-Through Estimator (STE) [11] enabled backpropagation through the sign function. BNNs are effective for discriminative tasks such as image classification [12], [13], but generative modelling—where the network must capture a full data distribution rather than assign labels—has seen far less exploration at the binary level.

The first attempt to fully binarize a diffusion model is BiDM [1], which operates at W1A1 precision and reports FID 22.74 on LSUN-Bedrooms at 256×256. BiDM combines a Timestep-friendly Binary Structure (TBS) with Space Patched Distillation (SPD), achieving a 28.0× storage reduction and 52.7× fewer operations, and reports that fully binarizing weights and activations without such machinery is catastrophic. A notable dependency is that BiDM relies on a pre-trained full-precision teacher for its distillation stage; the from-scratch approach we study here does not. We stress that BiDM operates at a far more demanding scale than our MNIST study, and we do not position our results as directly comparable to theirs.

D. Pre-Activation Residual Networks

He et al. [14] showed that moving batch normalization and the non-linearity ahead of the convolution (BN → Act → Conv) improves gradient propagation in deep residual networks. For binary layers the effect is useful for a related reason: because BatchNorm centres and scales the incoming activations toward zero mean and unit variance, downstream sign-based operations sit near their decision boundary, which we found kept STE gradients better behaved across training.

III. The Binarization Problem: Compression vs. Representation

A. A Hypothesis for Manifold Collapse

Inside a trained diffusion model the denoising function f_θ implicitly represents the score of the data distribution—the gradient $\nabla_x \log p_t(x)$ that pulls noisy samples back toward clean data. For any digit class this score field carves out manifold boundaries in pixel space, encoding stroke geometry and class identity in a high-dimensional directional structure.

PTQ replaces every floating-point weight $w_o \in \mathbb{R}$ with $\text{sign}(w_o) \cdot \alpha$, where the scale $\alpha = \text{mean}|W|$ attempts to compensate for lost magnitude. Because this replacement happens layer by layer after training, with no further gradient updates, the fine-grained magnitude ordering the network relied on is discarded. Our interpretation—which we offer as a hypothesis consistent with the measurements below, not as a proof—is that over the $T = 1{,}000$ reverse steps this incoherence compounds and degrades the sampled distribution, which is what we observe as the sharp FID increase under PTQ.

B. Native Binarization as a Representation Problem

Native Binarization flips the question around. Instead of squeezing a floating-point solution into binary constraints, we let the network *find* a binary solution from scratch. Starting from random initialization, gradient descent via the STE searches for a configuration of $\{-1, +1\}$ weights that directly minimizes the DDPM noise-prediction loss. The weights that emerge are not a degraded copy of a floating-point original; they are a binary encoding of the score field, expressed in a different representational geometry.

Our working hypothesis is that such a natively binary solution both exists and is reachable through standard gradient-based optimization on MNIST, given two conditions: (1) each binarization step preserves enough directional information to sustain a useful gradient signal (*structural dominance*), and (2) the gradient path through binary layers stays well-conditioned throughout training (*topological stability*). We do not claim these conditions are sufficient in general—only that they are consistent with what we observe here.

IV. Methodology

A. Native 1-Bit Diffusion Architecture

The backbone is a compact residual U-Net [15] with three channel widths—$[64, 128, 256]$—and two rounds of spatial down-sampling (MaxPool2d, $\times 2$) followed by matching up-sampling stages (nearest-neighbour, $\times 2$). Encoder feature maps are concatenated with their decoder counterparts via skip connections at every resolution. All experiments target 28×28 single-channel MNIST [16] images; the topology is shown in Figure 1.

Time Conditioning. Each diffusion step index is encoded with a 32-dimensional sinusoidal embedding [17] and fed through a small two-layer MLP, producing $t_{\text{emb}} \in \mathbb{R}^{32}$. Inside every residual block a learned linear layer maps t_{emb} to the block's channel count and adds it element-wise to the feature map, so timestep information is available at all spatial resolutions.

Boundary Layer Convention. Following standard BNN practice, the input convolution ($1\to 64$) and the output projection ($64\to 1$) remain in full precision across all variants. Keeping the pixel-space interface layers at higher precision avoids introducing quantization noise at the boundaries of the network.

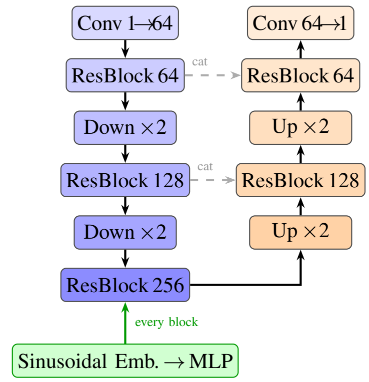


Fig. 1: Residual U-Net architecture shared across all three model variants (FP16, W1A16, W1A1). The encoder–decoder structure with channel widths [64, 128, 256], skip connections (dashed, via concatenation), and sinusoidal time embedding injection (green) is identical in all variants; only the convolution type changes (Conv2d / BitConv2d_Std / BitConv2d_BNN).

B. Pre-Activation Block Structure

Every binary residual block follows the pre-activation layout in Figure 2. A single block computes:

$$\begin{aligned} h &\leftarrow \text{BinaryAct}(\text{BN}(x)) \\ h &\leftarrow \text{BinaryConv2d}(h) \\ h &\leftarrow h + \text{Linear}(t_{\text{emb}}) \\ h &\leftarrow \text{BinaryAct}(\text{BN}(h)) \\ h &\leftarrow \text{BinaryConv2d}(h) \\ \text{output} &\leftarrow h + \text{skip}(x) \end{aligned}$$

where `skip` is a 1×1 binary convolution when channel dimensions change and an identity otherwise. In the W1A16 variant, `BinaryAct` is a full-precision activation (SiLU) and only the weights are binarized; in the W1A1 variant, `BinaryAct` is the clipped-STE sign activation described in Section IV-D.

Topological Stability via Pre-Activation Ordering. The motivation for this ordering is that BatchNorm re-centres and re-scales activations *before* they reach the following operation. Because the resulting distribution sits near zero mean with unit variance, sign-based operations stay close to their decision boundary during training. In our runs this ordering avoided the dead-neuron and gradient-starvation behaviour we observed with other orderings. We use **topological stability** as an informal name for this property: the binary representation stays balanced enough to keep training stable and to capture the class-level structure of the data on this benchmark.

C. Mean-Centered Weight Binarization (Structural Dominance)

Weight binarization in the W1A16 variant is handled by `BitConv2d_Std`, a convolution layer that subtracts the per-filter mean before applying the sign function. Given a weight

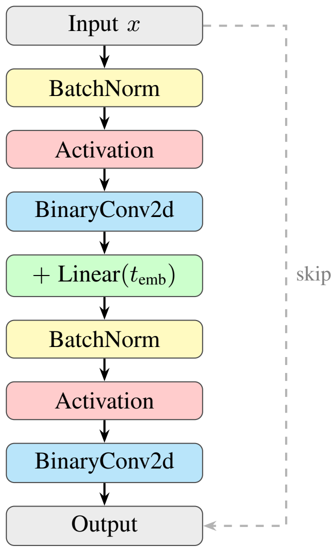


Fig. 2: Pre-activation binary residual block. BatchNorm is applied *before* the activation at both sub-layers, keeping the inputs to downstream binary operations approximately zero-centred during training (topological stability). The identity skip connection preserves gradient flow to shallow layers. The activation is SiLU for W1A16 (weights binarized only) and the clipped-STE sign function for W1A1 (weights and activations binarized).

tensor $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$, let $w_o \in \mathbb{R}^{C_{\text{in}} k^2}$ denote the flattened kernel for output channel o.

Standard (uncentered) binarization:

$$\tilde{w}_o = \operatorname{sign}(w_o) \cdot \alpha_o, \quad \alpha_o = \frac{\|w_o\|_1}{C_{\text{in}} \cdot k^2} \tag{2}$$

Centered binarization (`BitConv2d_Std`):

$$\bar{w}_o = w_o - \mu_o, \quad \mu_o = \operatorname{mean}(w_o) \tag{3}$$

$$\hat{w}_o = \operatorname{sign}(\bar{w}_o) \cdot \bar{\alpha}_o, \quad \bar{\alpha}_o = \frac{\|\bar{w}_o\|_1}{C_{\text{in}} \cdot k^2} \tag{4}$$

Structural Dominance Property. By subtracting μ_o, the isotropic DC component is removed from the weight vector, leaving its directional content. The sign function then quantizes this centred direction, while $\bar{\alpha}_o$ serves as a magnitude proxy that rescales the result. The intended effect is that $\hat{w}_o$ retains the *dominant directional structure* of the original filter w_o: a zero-mean vector's sign pattern reflects how its entries are distributed across directions more faithfully than that of a vector biased by a large DC offset, so centering reduces the angular distortion that binarization introduces. We expect this to matter for diffusion networks, whose convolutional kernels must capture oriented denoising responses to approximate the score field.

D. Gradient Approximation via Straight-Through Estimator

Both weight and activation binarization rely on the sign function, which has zero derivative almost everywhere. To enable backpropagation through this hard threshold we use the **Straight-Through Estimator** (STE) [11].

Weight STE (`BitConv2d_Std` and `BitConv2d_BNN`):

$$w_{\text{forward}} = \underbrace{(\hat{w}_o - w_o)}_{\text{stop-grad}} + w_o \tag{5}$$

Activation STE (W1A1 only, clipped):

$$\text{BinaryAct}(x) = \operatorname{sign}(x), \quad \frac{\partial\, \text{BinaryAct}}{\partial x} = \mathbf{1}[|x| \le 1] \tag{6}$$

Under the clipped variant [18], gradients are set to zero whenever $|x| > 1$, which suppresses the influence of outlier activations while keeping gradient information flowing through the bulk of the distribution.

V. Experimental Setup

A. Model Architectures and Initialization

Three model variants, all built on an identical U-Net backbone, are evaluated. Their configurations are summarized in Table I.

TABLE I: Model Variants

Variant	Weights	Acts.	Key Layer
ResUNet_FP16	FP16	FP16 (SiLU)	Conv2d
ResUNet_W1A16	1-bit, centered	FP16 (SiLU)	BitConv2d_Std
ResUNet_W1A1	1-bit, uncentered	1-bit (clip. STE)	BitConv2d_BNN

Every model starts from PyTorch's default initialization. We train with the standard DDPM objective under a linear noise schedule $\beta \in [10^{-4}, 0.02]$ spanning $T = 1{,}000$ diffusion steps. The FP16 and W1A16 variants use Adam (lr $= 10^{-3}$); W1A1 uses AdamW [19] (lr $= 10^{-4}$). All variants are trained on the full MNIST [16] training split (60,000 images, 28×28 grayscale, normalized to $[-1, 1]$). The time-embedding MLP uses ReLU for FP16 and W1A16 and GELU for W1A1. To reduce sampling variance, every reported metric is averaged over three independent evaluation runs (fresh sample draws) of the same trained checkpoint.

B. The W1A16 Isolation Strategy

Central to our design is a **controlled ablation** in which activation precision is held fixed at FP16 and only the weight-binarization strategy differs. This decouples the binarization methodology (PTQ versus native) from confounds introduced by activation-quantization noise. The two variants compared are:

- **ResUNet_W1A16 (Native):** Trained from scratch with `BitConv2d_Std`.
- **ResUNet_W1A16 (PTQ):** The trained FP16 model with weights post-hoc binarized via `BitConv2d_Std`. Biases and BatchNorm statistics are preserved; only the convolutional weights are binarized.

Because activation precision is identical in both settings, any gap in generation quality can be attributed to the training methodology—native versus post-training binarization—rather than to differences in activation representation.

C. Evaluation Metrics

Fréchet Inception Distance (FID). We use FID [20], which measures the distance between two multivariate Gaussians fit to InceptionV3 features—one from $N = 2{,}000$ generated samples, the other from the MNIST test set. Before feature extraction, all images are bilinearly upsampled from 28×28 to 299×299 and replicated across 3 channels:

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\Big) \quad (7)$$

We note that FID computed on MNIST with an ImageNet-trained InceptionV3 backbone is an imperfect proxy—the feature extractor was not trained on digits—so we treat FID here as a relative comparison across our own variants rather than an absolute quality figure.

Legibility Score. As a complementary, task-specific probe we report the Legibility Score: the average peak softmax confidence over $N = 1{,}000$ generated images, evaluated by a small MNIST classifier trained separately on real data:

$$\text{Legibility} = \frac{1}{N}\sum_{i=1}^{N} \max_{k\in\{0,\ldots,9\}} P\big(k \mid x_i^{\text{gen}}\big) \quad (8)$$

Values close to 1.0 indicate that generated digits commit strongly to a class; values near 0.10 (uniform chance) indicate near-collapse.

VI. Results and Analysis

A. Quantitative Comparison

The main results from the W1A16 ablation, together with the W1A1 variants, appear in Table II.

TABLE II: MNIST Benchmark: Native vs. Post-Training Quantization

Model	Strategy	FID ↓	Legibility ↑
ResUNet_FP16	Native (baseline)	29.60 ± 0.23	$90.79 \pm 0.51\%$
ResUNet_W1A16	**Native (ours)**	$\mathbf{24.24 \pm 1.15}$	$\mathbf{89.45 \pm 0.73\%}$
ResUNet_W1A16	PTQ	191.94 ± 1.37	$64.92 \pm 1.16\%$
ResUNet_W1A1	**Native (ours)**	$\mathbf{83.98 \pm 1.66}$	$\mathbf{81.02 \pm 0.54\%}$
ResUNet_W1A1	PTQ	223.00 ± 1.89	$63.04 \pm 0.74\%$

The pattern is consistent across metrics. With native training, the W1A16 model records FID 24.24 and legibility 89.45%, on par with the FP16 baseline (FID 29.60, legibility 90.79%). We deliberately avoid reading the small FID difference as evidence that the binary model is *superior*—given the single dataset, the imperfect FID proxy discussed above, and run-to-run variance, the honest summary is that native binary-weight training incurs no meaningful quality loss on MNIST. PTQ, by contrast, degrades sharply: FID 191.94 (roughly 7.9× the native model) with legibility falling to 64.92%, consistent with the view that retroactive binarization disrupts the learned score field, though some residual structure clearly persists. In the fully binary W1A1 regime, native training reaches FID 83.98 and 81.02% legibility—markedly worse than W1A16 but far from collapse—whereas PTQ decays further to FID 223.00 and 63.04%. The W1A16 contrast is visualised in Figure 3.

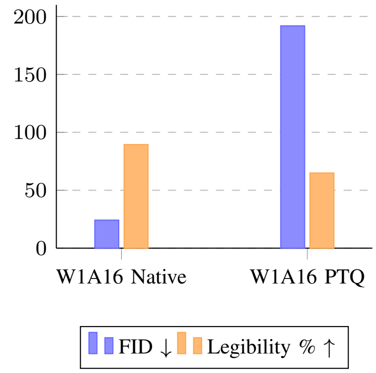


Fig. 3: FID (lower is better, blue) and Legibility % (higher is better, orange) for the two W1A16 strategies on MNIST. The native model attains markedly lower FID (24.24 vs. 191.94) and higher legibility (89.45% vs. 64.92%) than PTQ. Values are means over three evaluation runs.

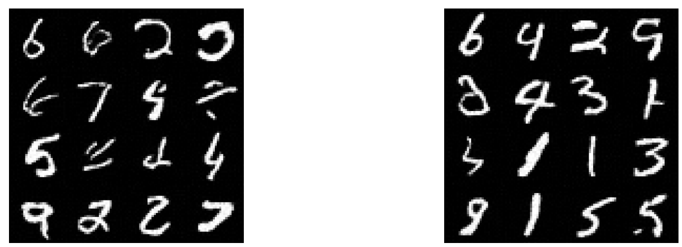


Fig. 4: ResUNet_FP16 (full-precision baseline). Generated digits are crisp, varied, and clearly committed to individual classes—the quality reference for the compressed variants.

B. Visual Evidence

Figures 4–9 provide visual corroboration of the numbers.

C. Hardware Efficiency (Theoretical)

Relative to ResUNet_FP16, the W1A16 model offers the following theoretical hardware advantages. We label these as theoretical because our current implementation uses standard PyTorch convolutions on binary-valued weight tensors and does not yet include specialised binary kernels.

- **Memory:** Each binarized convolutional weight requires only 1 bit versus 32 bits under FP32 storage—a **32× reduction** across the binarized layers (16× relative to FP16 inference weights).
- **Arithmetic:** Multiplication with a binary weight reduces to a sign flip, implementable as XNOR-popcount—a **58× reduction in binary operations** against the FP16 MAC count (BOPs vs. FLOPs).
- **Energy:** On dedicated binary-arithmetic hardware, such operations can draw roughly 1–2 orders of magnitude less energy than floating-point equivalents.

On MNIST these savings come at little measurable cost to class-level semantic quality: with 89.45% legibility versus 90.79% for FP16, the W1A16 model produces digits that are, at the class level, close to those of the full-precision baseline.

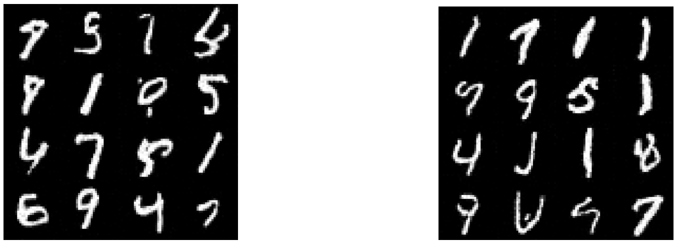


Fig. 5: Natively trained ResUNet_W1A16 (centered 1-bit weights, FP16 activations). Digit structure and class diversity remain visually intact, consistent with the measured FID of 24.24 and 89.45% legibility.

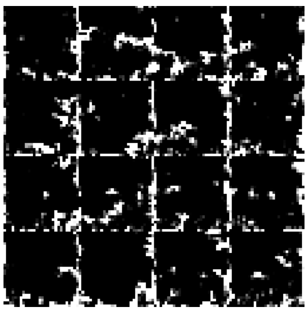


Fig. 6: PTQ ResUNet_W1A16 (FP16 model binarized after training). Outputs show pronounced structural degradation, matching the measured FID of 191.94 and 64.92% legibility, though some residual digit structure persists.

Fig. 7: Natively trained ResUNet_W1A1 (fully 1-bit: binary weights and activations, pre-activation ordering with clipped STE). Digit topology is largely preserved despite all-binary computation, though samples are visibly softer than the W1A16 model.

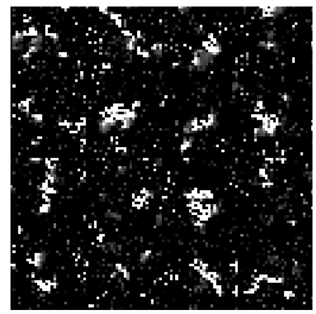


Fig. 8: PTQ ResUNet_W1A1 (FP16 model fully binarized in both weights and activations). Activation binarization compounds the weight-quantization error, producing further degradation beyond PTQ W1A16 (Fig. 6).

VII. Discussion

A. Why PTQ Fails at 1-Bit (on This Benchmark)

The structural-dominance view suggests a mechanism for why PTQ collapses at the 1-bit boundary. In a trained FP16 diffusion model, the score function is encoded through the joint magnitudes and signs of filter weights. Binarization discards the magnitude ordering and keeps only the signs; in a converged FP16 network, individual signs carry little standalone semantic content, so extracting them post-hoc removes most of the encoding.

A complementary explanation is landscape-theoretic. The loss surfaces of the DDPM objective under binary and full-precision parameterisations occupy largely disjoint regions: the minima reachable by $\{-1, +1\}$ weights need not coincide with those of the FP16 model. PTQ projects the FP16 optimum onto the binary parameter space, landing at a point that is neither optimal nor stationary under the binary loss, whereas native training explores the binary landscape from the outset. We offer both accounts as plausible interpretations consistent with our MNIST measurements rather than as established theory.

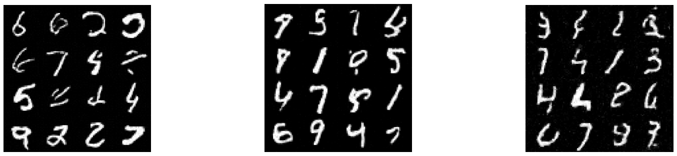
B. Relation to BiDM's Distillation

BiDM [1] reports FID 22.74 on LSUN-Bedrooms 256×256 at W1A1 precision using Timestep-friendly Binary Structure (TBS) and Space Patched Distillation (SPD). Its pipeline is distillation-dependent: the binary student is trained to mimic a full-precision teacher, so the binary model cannot be obtained without first training that teacher. Native Binarization, in the setting we study, is teacher-free—the binary network is trained directly. We are careful not to over-generalize this contrast: BiDM operates at a resolution and dataset far beyond MNIST, and our teacher-free result is demonstrated only in the much simpler regime studied here. The practical appeal of a teacher-free route—no pre-trained model required, a single binary model to train—is nonetheless worth noting for settings where teachers are unavailable or expensive.

We also note that the structural-dominance idea (centered weight binarization to retain directional filter structure) is architecturally orthogonal to BiDM's TBS and could in principle be combined with a learnable binarizer.

C. Legibility as a Complement to FID

Legibility captures something FID alone does not. FID rewards diversity and feature-space fidelity but is blind to whether any single sample commits to a class—and, as noted, its InceptionV3 backbone is a poor match for digits. A generator that produces varied but structurally ambiguous digits could attain a moderate FID yet score poorly on Legibility. Reporting both is more informative: FID gauges how well the generator spans the distribution, while Legibility tests whether individual samples are recognizable members of it.



(a) FP16 (b) W1A16 Native (c) W1A1 Native

Fig. 9: Three-way visual comparison. Quality degrades progressively with deeper quantization: the W1A16 native model preserves near-baseline fidelity, while W1A1 native shows the characteristic softening expected from fully binary activations.

D. Limitations

The most important limitation is scope: our experiments are confined to MNIST, whose manifold is far simpler and lower-dimensional than that of natural images. We view this study as a first probe, and scaling Native Binarization to CIFAR-10, CelebA, or LSUN—where the score field is substantially more complex—is essential future work and may well reveal failure modes that MNIST does not expose. Additional limitations: the Legibility Score depends on a task-specific classifier and does not transfer to continuous-domain generation; FID on MNIST relies on an ImageNet-trained feature extractor and is best read as a relative measure; the $58\times$ operational speedup is theoretical and would require specialised XNOR-popcount kernels to realise; and the "structural dominance" and "topological stability" principles are empirical design intuitions rather than formally proven properties.

VIII. Conclusion

We presented **Native Binarization**, a from-scratch training recipe for 1-bit-weight diffusion models, and studied it on MNIST. Its two ingredients are mean-centered weight binarization (*structural dominance*) and pre-activation residual ordering (*topological stability*). In a controlled W1A16 ablation, the sharp quality loss under standard PTQ (FID 191.94, legibility 64.92%) did not appear when binary weights were trained from scratch: the native W1A16 model reached FID 24.24 and 89.45% legibility, comparable to an FP16 baseline of identical architecture, and a fully binary W1A1 variant degraded gracefully rather than collapsing.

We present these as preliminary, MNIST-scale findings rather than a general resolution of binary diffusion. Within that scope, they offer a concrete data point suggesting that the collapse often attributed to 1-bit precision is, at least here, more closely tied to the train-then-quantize pipeline than to binarization itself—and that a teacher-free, from-scratch route is worth studying further. Whether structural dominance and topological stability continue to help at natural-image scale, and across other generative families such as flow matching, consistency, and masked diffusion models, is an open question we hope to address in future work.

Code Availability. Source code and trained checkpoints are publicly available at https://github.com/nihal-gazi/Native-Binarization.

References

[1] X. Zheng, X. Liu, Y. Bian, X. Ma, Y. Zhang, J. Wang, J. Guo, and H. Qin, "BiDM: Pushing the limit of quantization for diffusion models," in *Advances in Neural Information Processing Systems*, 2024, arXiv:2412.05926.

[2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 6840–6851.

[3] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021.

[4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695.

[5] D. Podell, Z. English, K. Lacey, A. Blattmann, T. Dockhorn, J. Müller, J. Penna, and R. Rombach, "SDXL: Improving latent diffusion models for high-resolution image synthesis," *arXiv preprint arXiv:2307.01952*, 2023.

[6] T. Salimans and J. Ho, "Progressive distillation for fast sampling of diffusion models," in *International Conference on Learning Representations*, 2022.

[7] X. Li, Y. Liu, L. Lian, H. Yang, Z. Dong, D. Kang, S. Zhang, and K. Keutzer, "Q-diffusion: Quantizing diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 17 535–17 545.

[8] Y. He, L. Liu, J. Liu, W. Wu, H. Zhou, and B. Zhuang, "PTQD: Accurate post-training quantization for diffusion models," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[9] I. Hubara, M. Courbariaux, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[10] M. Rastegari, V. Ordonez, J. Redmon, and A. Farhadi, "XNOR-Net: ImageNet classification using binary convolutional neural networks," in *European Conference on Computer Vision*. Springer, 2016, pp. 525–542.

[11] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," *arXiv preprint arXiv:1308.3432*, 2013.

[12] Z. Liu, Z. Shen, M. Savvides, and K.-T. Cheng, "ReActNet: Towards precise binary neural network with generalised activation functions," in *European Conference on Computer Vision*. Springer, 2020, pp. 143–159.

[13] B. Martinez, J. Yang, A. Bulat, and G. Tzimiropoulos, "Training binary neural networks with real-to-binary convolutions," in *International Conference on Learning Representations*, 2020.

[14] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *European Conference on Computer Vision*. Springer, 2016, pp. 630–645.

[15] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2015, pp. 234–241.

[16] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, source of sinusoidal positional embeddings used for time conditioning.

[18] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to $+1$ or -1," *arXiv preprint arXiv:1602.02830*, 2016, introduces the clipped Straight-Through Estimator for binary activation training.

[19] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations*, 2019.

[20] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, introduces the Fréchet Inception Distance (FID).